

OILY SKIN AS A PHYSIOLOGICAL RESPONSE TO HAMER LOOP ACTIVITY

*Seborrhoea as a Distal Consequence of Systemic Inflammation Rather than a
Primary Cause*

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Abstract

This document, the fifth in the Hamer Loop Manuscript Series, addresses the role of seborrhoea — oily skin — within the Hamer Loop framework. Conventional dermatological models treat excess sebum production as an upstream causal factor in acne and rosacea: oil clogs follicles, creates an environment for bacterial proliferation, and lesions form as a consequence. The Hamer Loop framework inverts this relationship entirely.

We propose that seborrhoea is not a cause of inflammatory skin disease in susceptible individuals but a distal physiological response to it — a downstream consequence of the systemic immune activity initiated by dietary Neu5Gc exposure. Four sequential mechanisms are proposed: histamine-induced vasodilation and increased facial blood flow; thermal stimulation of sebaceous glands by the resulting localised temperature rise; mechanical displacement of existing sebum by elevated interstitial pressure during nocturnal amplification; and, more speculatively, activation of a secondary clearance pathway in which the sebaceous system expresses lipid-soluble inflammatory mediators when lymphatic drainage is functionally impaired.

The primary observational evidence for this reframing is direct and reproducible: during deliberate elimination trials, oily skin resolved completely within the same 7–10 day window as acne lesions, rosacea, peripheral oedema, and gastrointestinal symptoms — without any topical or pharmacological sebosuppressive intervention. Its co-resolution with conditions that have no sebaceous component whatsoever is incompatible with a model in which sebum overproduction is an autonomous primary pathology.

Keywords: *seborrhoea; oily skin; sebaceous gland; Hamer Loop; xenosialitis; histamine; vasodilation; interstitial pressure; nocturnal amplification; lymphatic stasis; dietary elimination; acne; rosacea*

1. The Central Argument: Inverting the Causal Arrow

The conventional understanding of acne pathogenesis positions sebum overproduction as an upstream event in a causal chain: excess oil accumulates in the follicular canal, creates an anaerobic environment favourable to *Cutibacterium acnes* proliferation, triggers an inflammatory response to bacterial products, and lesions form as the result. In this model, treating sebum production directly — through retinoids, sebosuppressants, and topical agents targeting the sebaceous gland — is a rational first-line approach.

The Hamer Loop framework does not dispute that this sequence occurs. Document 2 of this series explicitly frames Category 2 and Category 3 lesions as the handoff phase in which conventional sebaceous and microbial pathophysiology takes over from the initiating lymphatic-immune phase. The bacterial colonisation is real. The sebaceous involvement is real.

What the Hamer Loop framework disputes is the direction of causality at the origin of the process. In susceptible individuals, the initiating event is not sebum overproduction — it is dietary Neu5Gc exposure, immune recognition, mast cell degranulation, and the inflammatory cascade that follows. The sebum overproduction is what the skin does in response to that cascade. It is the consequence, not the cause. Treating it as the cause while leaving the initiating dietary trigger in place is the equivalent of treating smoke inhalation by opening a window while leaving the fire burning.

2. Four Mechanisms of Seborrhoea in the Hamer Loop

Four sequential mechanisms are proposed by which Hamer Loop activity produces excess sebum production and surface oiliness. The first two are mechanistically grounded in established physiology. The third is a direct extension of the nocturnal amplification mechanism documented in Document 1. The fourth is explicitly speculative and is offered as a hypothesis for investigation rather than an established finding.

Mechanism	Proposed Pathway	Epistemic Status
1. Histamine-induced vasodilation	Mast cell degranulation → histamine release → vasodilation → ↑ facial blood flow → ↑ local skin temperature	Established physiology; proposed extension to sebaceous response
2. Thermal sebaceous stimulation	↑ local temperature → sebaceous gland upregulation → ↑ sebum synthesis and secretion as thermoregulatory response	Established sebaceous physiology; mechanistically coherent extension
3. Mechanical interstitial displacement	Nocturnal fluid redistribution → ↑ interstitial pressure → mechanical compression of pilosebaceous units → sebum displaced to surface	Direct extension of nocturnal amplification mechanism (Doc 1, Section 5)
4. Secondary lymphatic clearance	Lymphatic functional impairment → sebaceous system activated as alternative route for lipid-soluble mediator expression	Explicitly speculative; offered as hypothesis for investigation

Table 1. Four proposed mechanisms of seborrhoea within the Hamer Loop framework, with epistemic status clearly distinguished. Mechanisms 1–3 are mechanistically grounded; Mechanism 4 is speculative and acknowledged as such throughout this document.

3. Mechanism 1: Histamine-Induced Vasodilation

The mechanistic sequence begins at Stage 1 of the Hamer Loop: immune recognition of incorporated Neu5Gc by circulating anti-Neu5Gc antibodies triggers mast cell degranulation and the rapid release of histamine. As documented in Table 1 of Document 1, this occurs within minutes of trigger ingestion and manifests as rhinorrhoea, flushing, and vasomotor activation.

Histamine induces rapid vasodilation — widening of blood vessels — and increases capillary permeability, allowing fluid and inflammatory mediators to shift into interstitial compartments. In the face and upper head-neck territory, where Hamer Loop lesion presentations are most pronounced, this vasodilation produces a marked and rapid increase in cutaneous blood flow. The flushing that susceptible individuals observe in the mirror within minutes of consuming a trigger food is the visible surface manifestation of this vascular event.

This increased facial blood flow is not merely a symptom in itself. It is the initiating event for the sebaceous response described in the following section. The vascular opening that histamine produces creates the thermal conditions that the sebaceous gland subsequently responds to.

4. Mechanism 2: Thermal Stimulation of Sebaceous Glands

Sebaceous glands are highly sensitive to local temperature. Elevated skin temperature is a recognised physiological trigger for increased sebum synthesis and secretion — a relationship established in the dermatological literature and consistent with the well-documented observation that sebum production increases in warmer climates and warmer seasons.

4.1 The Thermal Response

As Hamer Loop activity intensifies and cutaneous blood flow increases under histamine-mediated vasodilation, local skin temperature rises measurably in the face and scalp territories most densely vascularised and most heavily targeted by TRM cell activity. Sebaceous glands in these territories respond to the thermal stimulus by upregulating sebum production — a thermoregulatory mechanism by which the skin attempts to dissipate the localised thermal load through the insulating and evaporative properties of the sebaceous film.

The surface oiliness that susceptible individuals observe as a characteristic feature of an active flare is therefore proposed to be, in significant part, a thermoregulatory response to the inflammatory vasodilation of Stage 1 — not a primary sebaceous pathology. The sebaceous

gland is doing exactly what it is designed to do in response to elevated local temperature. The problem is not the gland's response; it is the immune cascade that is elevating the temperature in the first place.

4.2 The Conventional Misattribution

The conventional dermatological model observes elevated sebum production in acne patients and reasonably concludes that this overproduction is driving the condition. Within the Hamer Loop framework, this is a misattribution of the causal sequence. The sebaceous gland is a responder, not an initiator. Targeting it with sebosuppressive agents — retinoids, isotretinoin, hormonal therapies — addresses the thermoregulatory response while leaving the Hamer Loop mechanism, and the dietary trigger that drives it, entirely unaddressed. Symptom management through sebosuppression is a rational palliative approach; it becomes a therapeutic dead end when the initiating mechanism is not simultaneously addressed.

5. Mechanism 3: Mechanical Interstitial Displacement

A distinct and mechanistically independent contribution to surface oiliness arises from the nocturnal amplification mechanism documented in Section 5 of Document 1. This mechanism does not involve increased sebum synthesis; it involves the physical displacement of existing sebum under elevated dermal pressure.

5.1 The Pressure Displacement Mechanism

As the Hamer Loop progresses into its Stage 4 nocturnal amplification phase, recumbent fluid redistribution toward the head and neck territory increases interstitial pressure within the dermis. The pilosebaceous unit — the follicular structure that houses both the hair shaft and the sebaceous gland — is surrounded by this pressurised interstitial fluid. Rising pressure compresses the structures immediately surrounding the follicular opening and mechanically displaces existing sebum toward the skin surface.

The result is the sudden and pronounced surface oiliness that susceptible individuals characteristically observe upon waking. This is not new sebum production; it is the overnight mechanical expression of sebum that was already present in the follicular canal, driven outward by the elevated interstitial pressure of the nocturnal amplification phase. The timing — maximal oiliness in the morning, correlating with the overnight fluid redistribution — is itself evidence for this mechanical rather than purely secretory origin.

5.2 Additive Effect with Thermal Stimulation

Mechanisms 2 and 3 are distinct in their origin — one secretory, one mechanical — but additive in their effect. The thermal stimulation of sebaceous glands during the active daytime Hamer Loop phase increases the total sebum available in the follicular canal. The nocturnal interstitial pressure then expresses this accumulated sebum to the surface during sleep. The morning presentation reflects the combined output of both mechanisms, which is why the surface oiliness in susceptible individuals during an active flare is characteristically most pronounced

upon waking and tends to diminish over the course of the day as the mechanical displacement effect resolves.

6. Mechanism 4: Secondary Lymphatic Clearance (Speculative)

The fourth proposed mechanism is explicitly and deliberately speculative. It is included because it is mechanistically coherent, because it provides a potential explanation for the persistent low-grade oiliness observed between acute flares rather than only during them, and because it represents a testable hypothesis that, if validated, would substantially deepen the understanding of sebaceous function in inflammatory skin disease.

The Hamer Loop model proposes that regional lymphatic flow becomes functionally limited during active inflammatory cycles, preventing efficient clearance of the lipid-soluble inflammatory mediators that accumulate in the interstitial reservoir. The hypothesis proposed here is that when the lymphatic system is functionally overwhelmed, the sebaceous system may be recruited as an alternative clearance pathway — a secondary route through which lipid-soluble compounds can be expressed to the body surface rather than remaining sequestered in the interstitial space.

This is analogous, in principle, to the skin's established role as an organ of excretion for lipid-soluble compounds — a function well-documented for certain environmental toxins, pharmaceutical metabolites, and aromatic compounds. Whether the specific class of inflammatory mediators produced by Neu5Gc xenosialitis are expressed via this route has not been tested. The hypothesis is offered as a direction for investigation, not a claim of established mechanism.

If confirmed, this mechanism would explain a specific observational pattern: the persistent, low-grade oiliness reported by susceptible individuals even in the intervals between acute flares, which is difficult to account for by thermal stimulation alone (which requires an active vasodilation event) or by mechanical displacement alone (which requires nocturnal pressure elevation). A baseline elevation in sebaceous output between flares would be consistent with a continuous, low-level secondary clearance function operating in parallel with the episodic flare-associated mechanisms.

7. The Elimination Trial: Primary Observational Evidence

The strongest evidence for the proposed reframing of seborrhoea as consequence rather than cause is not mechanistic — it is observational, and it is direct.

During deliberate elimination trials in which all primary dietary triggers — bovine-derived foods, wheat, and dairy derivatives — were simultaneously removed, oily skin resolved completely within the same 7–10 day window as acne lesions, rosacea, peripheral oedema, joint pain, and gastrointestinal symptoms. The full systemic resolution profile is documented in Section 6.4 of

Document 1. The critical observation for the purposes of this document is the temporal and categorical behaviour of the seborrhoea within that profile.

7.1 Co-Resolution as Evidence

Oily skin did not resolve after the lesions cleared. It did not resolve more slowly, reflecting a hypothetical lag between inflammation resolution and sebaceous normalisation. It resolved within the same window, at the same rate, as conditions that have no sebaceous component whatsoever — peripheral oedema, joint pain, nasal congestion, and sleep apnoea.

This temporal co-resolution is incompatible with a model in which sebum overproduction is an autonomous primary pathology. If the sebaceous gland were the independent driver of the skin condition, its dysfunction would persist beyond the resolution of the immune-mediated components, or at minimum show a different resolution trajectory. It does not. It resolves in lockstep with the rest of the systemic inflammatory picture — consistent only with a model in which it is driven by the same upstream mechanism as every other symptom in the profile.

7.2 Re-Exposure Confirmation

Upon reintroduction of dietary triggers following the elimination period, oily skin returned alongside the cutaneous and systemic symptoms — within the same temporal window and following the same Stage 1 to Stage 4 progression documented in the Hamer Loop timeline. Its return as a dependent variable within the Hamer Loop cycle, rather than as a persistent background condition independent of dietary exposure, further confirms its position as a consequence rather than a cause.

No topical sebosuppressive intervention was used during these elimination and reintroduction trials. The resolution and return of seborrhoea were produced entirely by the presence or absence of the dietary trigger. This is the most direct possible evidence that sebum overproduction in this subject is an upstream-triggered response rather than an autonomous sebaceous malfunction.

8. Clinical and Therapeutic Implications

If the proposed reframing is accepted — or even taken seriously as a hypothesis warranting investigation — it has immediate implications for the clinical management of seborrhoea-associated inflammatory skin conditions.

8.1 The Limitation of Sebosuppressive Monotherapy

Isotretinoin, the most powerful sebosuppressive agent in clinical use, achieves dramatic reductions in sebum production and produces sustained remission in a significant proportion of acne patients. Within the Hamer Loop framework, this efficacy is not surprising: by suppressing sebum production, isotretinoin interrupts the sebaceous phase of the Hamer Loop cascade, reducing the substrate available for follicular occlusion and bacterial colonisation.

What it does not do is address the Neu5Gc-driven xenosialitis that initiates the cascade. The vascular response, the interstitial reservoir, and the TRM cell amplification mechanism remain active. This is consistent with the well-documented relapse pattern following isotretinoin discontinuation in patients who return to trigger-rich diets: the sebaceous suppression was not treating the cause; it was managing one of the consequences.

8.2 Dietary Elimination as a Complementary Strategy

The elimination trial data suggest that complete dietary trigger avoidance produces full resolution of seborrhoea without any sebosuppressive intervention. For susceptible individuals, this represents a mechanistically coherent alternative to pharmacological management — one that addresses the initiating cause rather than one of its consequences. The Acnerase platform is designed to make this dietary avoidance practically achievable by identifying hidden Neu5Gc sources and ammonium catalysts in food, beverages, and medications before consumption.

This is not an argument against pharmacological treatment. For individuals who cannot or choose not to pursue complete dietary elimination, sebosuppressive therapies remain valid and effective management tools. The Hamer Loop framework positions dietary modification and pharmacological sebosuppression as complementary strategies rather than competing ones — with the dietary approach addressing the initiating mechanism and the pharmacological approach managing the sebaceous consequence.

9. Limitations

The four mechanisms proposed in this document vary substantially in their evidential basis and should be treated accordingly. Mechanisms 1 and 2 — histamine-induced vasodilation and thermal sebaceous stimulation — draw on established physiology and represent mechanistically coherent extensions to the Hamer Loop framework. The specific claim that Hamer Loop-induced temperature changes are of sufficient magnitude and duration to drive clinically significant sebum upregulation has not been formally measured.

Mechanism 3 — mechanical interstitial displacement during nocturnal amplification — is a direct extension of the nocturnal amplification mechanism documented in Document 1. The magnitude of the interstitial pressure increase, and its capacity to produce measurable sebum displacement in the pilosebaceous unit, has not been quantified.

Mechanism 4 — the secondary lymphatic clearance hypothesis — is explicitly speculative and is acknowledged as such throughout this document. The proposal that the sebaceous system serves as an alternative expression route for lipid-soluble inflammatory mediators under conditions of lymphatic stasis is mechanistically plausible but entirely unvalidated. It is offered as a direction for investigation.

The observational evidence in Section 7 is the strongest component of this document. However, it is derived from a single subject and cannot exclude alternative explanations — including the possibility that the dietary changes responsible for Hamer Loop resolution also independently normalise sebaceous function through mechanisms unrelated to the proposed pathway (for

example, through changes in insulin-like growth factor 1 signalling, which is known to influence sebaceous activity and which may be modulated by the dietary changes involved in the elimination protocol).

10. Conclusion

The conventional model of acne pathogenesis positions the sebaceous gland as a malfunctioning upstream driver whose excess oil output sets the condition in motion. The Hamer Loop framework repositions it as a faithful, well-functioning downstream responder — one that is doing precisely what a sebaceous gland is supposed to do in response to the signals it receives from an inflamed and thermally elevated cutaneous environment.

The gland is not broken. The signal is wrong. And the wrong signal originates not in the skin but at the dinner table.

This reframing matters clinically because it changes where the therapeutic intervention should logically be directed. Targeting the sebaceous gland while leaving the dietary trigger in place is addressing the thermometer rather than the fever. It may make the thermometer read lower — isotretinoin undeniably does — but the underlying temperature remains elevated.

The elimination trial evidence is unambiguous on this point: remove the dietary trigger, and oily skin resolves within 7–10 days without any intervention directed at the sebaceous gland itself. The gland was never the problem. It was the canary. And like all canaries, what it was telling us was something about the mine.

References

- Pappas, A. (2009). The relationship of diet and acne: A review. *Dermato-endocrinology*, 1(5), 262–267.
- Picardo, M., Ottaviani, M., Camera, E., & Mastrofrancesco, A. (2009). Sebaceous gland lipids. *Dermato-endocrinology*, 1(2), 68–71.
- Thiboutot, D. M. (2004). Acne: hormonal concepts and therapy. *Clinics in Dermatology*, 22(5), 419–428.
- Melnik, B. C., & Schmitz, G. (2009). Role of insulin, insulin-like growth factor-1, hyperglycaemic food and milk consumption in the pathogenesis of acne vulgaris. *Experimental Dermatology*, 18(10), 833–841.
- Dhar, C., Sasmal, A., & Varki, A. (2019). From 'Serum Sickness' to 'Xenosialitis'. *Frontiers in Immunology*.
- Samraj, A. N., Pearce, O. M., et al. (2015). A red meat-derived glycan promotes inflammation and cancer progression. *Proceedings of the National Academy of Sciences*, 112(2), 542–547.

Hamer, M. (2026). Bioavailability Amplifiers: Ammonium Compounds and Lesion Phase Classification [Document 2 of the Hamer Loop Manuscript Series]. Hamer Glycan Systems [Unpublished].

Hamer, M. (1981–2026). Longitudinal bedside assessment and pattern recognition. Internal Case Files [Unpublished].